

Elamaran M

Full Stack Developer

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Career Summary

Deeply curious and execution-focused MERN Stack Developer, consistently driven to explore new technologies and bring innovative ideas to improve existing systems. Disciplined and determined to gain practical experience while turning complex ideas into working software.

Projects

DevMind

AI Agent for Developer Debugging

- Built a multi-tool **AI agent** using the **ReAct pattern** where LLM autonomously decides to read GitHub code or search the web, returning exact bug fixes with file name and line number.
- Integrated **Groq tool calling** with two custom tools — `read_github_file` (GitHub REST API) and `search_web` (Tavily API) — where LLM dynamically selects which tool to invoke instead of following hardcoded logic.
- Leveraged **openai/gpt-oss-120b** via Groq inference API with deterministic settings to ensure reliable multi-step tool calling across the agent loop.
- Secured per-project GitHub token storage with **JWT HTTP-only cookies** and protected routes, ensuring each user's repo and notes are fully isolated in MongoDB.
- Built **three-panel workspace UI** with file tree, agent chat, and auto-saved solutions panel using React and Redux Toolkit for global state management.

Nova-Share

AI-Powered File Management System

- Developed a full-stack secure file sharing platform with **JWT authentication**, protected routes, and HTTP-only cookies, ensuring end-to-end security.
- Integrated **Groq's Llama 3.3 70B LLM** to enable AI-powered smart file search and file summarization, demonstrating practical LLM integration.
- Utilized **Cloudinary API** for cloud-based file storage, enabling unique short link sharing, password protection, and configurable file expiry.
- Built a real-time dashboard tracking uploads, downloads, and file statistics using **Redux Toolkit** for state management and MongoDB for persistent storage.
- Deployed frontend on **Vercel** and backend on **Render** with environment-based configuration, CORS handling.

DailyDesk

RAG-based Personal Journal Assistant

- Built a full-stack MERN digital journal app with **RAG pipeline** using Qdrant vector database and OpenRouter LLM for semantic journal querying.
- Implemented **JWT authentication** with secure HTTP-only cookies, server-side validation, and protected routes for user data privacy.
- Integrated **Redux Toolkit (RTK Query)** for global state management and optimized REST API data fetching with automatic cache invalidation.
- Deployed React.js frontend on Vercel and Node.js/Express backend on Render, styled with **Tailwind CSS** and Daisy UI with full CI/CD via GitHub.

Skills

Frontend: React, JavaScript (ES6+), Tailwind CSS, HTML, CSS

Backend & Database: Node.js, Express.js, MongoDB, Mongoose, JWT, Qdrant (VectorDB)

Tools & Deployment: Git & Github, Vercel, Render

Additional Skills: OpenRouter, Prompt Engineering

Education

Government College of Engineering – Salem
BE in Mechanical Engineering

2022–2026
*CGPA – 7.63**